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HUBER
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in-vivo_20260706
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\in-vivo_20260706\localizer

TA: 15 sec Coil Selection: Manual Voxel Size: 0.5×0.5×5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	8.6 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None

Contrast - Common

Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter

Geometry - Common

Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up

System - Adjustments

B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	297.136349 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline - MapIt

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	qfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	AC

Sequence - Assistant

SAR Assistant	Off

Sequence - Assistant

Allowed Delay	0 s

\\MARTINOS DEVELOPER\HUBER\Physics_playground\in-vivo_20260706\wip_ShimToolbox_B0map_3mm

TA: 31 sec Coil Selection: Manual Voxel Size: 3.0×3.0×3.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.8 A21.6 F24.1 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	15.0 ms
TE 1	2.20 ms
TE 2	5.40 ms
Concatenations	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	15.0 ms
TE 1	2.20 ms
TE 2	5.40 ms
MTC	Off
Magn. Preparation	None
Flip Angle	8 deg
Fat-Water Contrast	Water Excitation
Dark Blood	Off
Contrasts	2
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	72
Phase Resolution	100 %
Slice Resolution	100 %

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.8 A21.6 F24.1 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	15.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.8 A21.6 F24.1 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.8 A21.6 F24.1
L	0.8 mm
A	21.6 mm
F	24.1 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.8 A21.6 F24.1 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	216 mm
F >> H	216 mm
R >> L	156 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.136349 MHz

System - Tx/Rx

? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	15.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Water Excitation
Magn. Preparation	None
Dark Blood	Off
FOV Read	216 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	2
TE 1	2.20 ms
TE 2	5.40 ms
TR	15.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	WIP_GRE
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Readout Mode	Monopolar
Gradient Mode	Whisper
Flow Compensation 1	None
Flow Compensation 2	None
Bandwidth 1	500 Hz/Px
Bandwidth 2	500 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	AC

Sequence - Special

Mode	B0 Mapping
Masking	Brain Extraction
Phase Unwrapping	Spatial

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\MARTINOS DEVELOPER\HUBER\Physics_playground\in-vivo_20260706\large_fov24

TA: 16 sec Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 4 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	1236 ms
TE 1	19.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	51.5 ms
Vol. TR	1236 ms
TE 1	19.00 ms
Multi-echo spacing	40.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	206
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	1
Reference Lines 3D	18
Reordering Shift 3D	0
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	1236 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm

Geometry - AutoAlign

A	20.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R3.6 A21.4 H0.6 mm
! Orientation	T > S0.4 > C-0.1
! Rotation	89.76 deg
! R >> L	151 mm
! A >> P	202 mm
! F >> H	42 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136349 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.100

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1104 Hz/Px
Echo Spacing	1.01 ms
Turbo Factor	24
Segmentation	1
EPI Factor	39

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	integrated
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 /10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Regular RO

Sequence - Assistant

SAR Assistant	Off

\\MARTINOS DEVELOPER\HUBER\Physics_playground\in-vivo_20260706\large_fov24_useme_control

TA: 10:22 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 4 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F2.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	1236 ms
TE 1	19.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	51.5 ms
Vol. TR	1236 ms
TE 1	19.00 ms
Multi-echo spacing	40.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	500
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	206
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	1
Reference Lines 3D	18
Reordering Shift 3D	0
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F2.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	1236 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F2.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F2.0
L	0.0 mm

Geometry - AutoAlign

A	20.0 mm
F	2.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R3.6 A21.4 H0.6 mm
! Orientation	T > S0.4 > C-0.1
! Rotation	89.76 deg
! R >> L	151 mm
! A >> P	202 mm
! F >> H	42 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136349 MHz
! Ref. Amplitude 1H	350.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.100

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1104 Hz/Px
Echo Spacing	1.01 ms
Turbo Factor	24
Segmentation	1
EPI Factor	39

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	integrated
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 /10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off

\\MARTINOS DEVELOPER\HUBER\Physics_playground\in-vivo_20260706\large_fov24_useme_ACDC

TA: 10:22 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 4 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F2.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	1236 ms
TE 1	19.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	51.5 ms
Vol. TR	1236 ms
TE 1	19.00 ms
Multi-echo spacing	40.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	500
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	206
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	1
Reference Lines 3D	18
Reordering Shift 3D	0
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F2.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	1236 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F2.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F2.0
L	0.0 mm

Geometry - AutoAlign

A	20.0 mm
F	2.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R3.6 A21.4 H0.6 mm
! Orientation	T > S0.4 > C-0.1
! Rotation	89.76 deg
! R >> L	151 mm
! A >> P	202 mm
! F >> H	42 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136349 MHz
! Ref. Amplitude 1H	350.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.100

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1104 Hz/Px
Echo Spacing	1.01 ms
Turbo Factor	24
Segmentation	1
EPI Factor	39

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	integrated
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 /10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off

\\MARTINOS DEVELOPER\HUBER\Physics_playground\in-vivo_20260706\large_fov24_useme_control_2

TA: 10:22 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 4 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F2.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	1236 ms
TE 1	19.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	51.5 ms
Vol. TR	1236 ms
TE 1	19.00 ms
Multi-echo spacing	40.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	500
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	206
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	1
Reference Lines 3D	18
Reordering Shift 3D	0
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F2.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	1236 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F2.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F2.0
L	0.0 mm

Geometry - AutoAlign

A	20.0 mm
F	2.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R3.6 A21.4 H0.6 mm
! Orientation	T > S0.4 > C-0.1
! Rotation	89.76 deg
! R >> L	151 mm
! A >> P	202 mm
! F >> H	42 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136349 MHz
! Ref. Amplitude 1H	350.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.100

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1104 Hz/Px
Echo Spacing	1.01 ms
Turbo Factor	24
Segmentation	1
EPI Factor	39

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	integrated
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 /10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off

\\MARTINOS DEVELOPER\HUBER\Physics_playground\in-vivo_20260706\large_fov24_useme_control_real

TA: 10:22 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 4 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F2.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	1236 ms
TE 1	19.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	51.5 ms
Vol. TR	1236 ms
TE 1	19.00 ms
Multi-echo spacing	40.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard

Contrast - Dynamic

Measurements	500
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	206
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	1
Reference Lines 3D	18
Reordering Shift 3D	0
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 F2.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	1236 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 F2.0 mm

Geometry - AutoAlign

Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 F2.0
L	0.0 mm
A	20.0 mm
F	2.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	R3.6 A21.4 H0.6 mm
! Orientation	T > S0.4 > C-0.1
! Rotation	89.76 deg
! R >> L	151 mm
! A >> P	202 mm
! F >> H	42 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136349 MHz
! Ref. Amplitude 1H	350.000 V
Reset	Off
Correction Factor	1.00

System - Tx/Rx

Image Scaling	0.100

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1104 Hz/Px
Echo Spacing	1.01 ms
Turbo Factor	24
Segmentation	1
EPI Factor	39

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	integrated
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 /10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off